Tutorial 8: Principal Component Analysis

Four problems in PCA theory and practice

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Task 1: PCA from sensor readings

Four readings from two correlated sensors are:

$$\boldsymbol{x}_1 = \begin{bmatrix} 13 \\ 21 \end{bmatrix}, \quad \boldsymbol{x}_2 = \begin{bmatrix} 11 \\ 23 \end{bmatrix}, \quad \boldsymbol{x}_3 = \begin{bmatrix} 7 \\ 19 \end{bmatrix}, \quad \boldsymbol{x}_4 = \begin{bmatrix} 9 \\ 17 \end{bmatrix}.$$

Use the population-style covariance convention for this tutorial:

$$\boldsymbol{S} = \frac{1}{n} \sum_{i=1}^{n} (\boldsymbol{x}_i - \bar{\boldsymbol{x}})(\boldsymbol{x}_i - \bar{\boldsymbol{x}})^{\mathsf{T}}, \qquad n = 4.$$

(a) Compute the mean and all four centered observations.

(b) Compute the covariance matrix.

(c) Derive the eigenvalues and unit eigenvectors. Identify PC1.

(d) Compute the PC1 explained-variance ratio and the PC1 coordinate of $\boldsymbol{x}_1$.

Solution 1(a): Centering

(a) Compute the training mean and centered readings. The two feature means are:

$$\bar{\boldsymbol{x}} = \frac{1}{4}\begin{bmatrix} 13+11+7+9 \\ 21+23+19+17 \end{bmatrix} = \boxed{\begin{bmatrix} 10 \\ 20 \end{bmatrix}}.$$

Subtract the same mean from every observation:

$$\boxed{\begin{aligned} \tilde{\boldsymbol{x}}_1 &= (3,1)^\mathsf{T}, & \tilde{\boldsymbol{x}}_2 &= (1,3)^\mathsf{T}, \\ \tilde{\boldsymbol{x}}_3 &= (-3,-1)^\mathsf{T}, & \tilde{\boldsymbol{x}}_4 &= (-1,-3)^\mathsf{T}. \end{aligned}}$$

The centered readings sum to $(0,0)^\mathsf{T}$, confirming that centering removes the feature means.

Solution 1(b): Covariance

(b) Compute the covariance matrix. Using the centered readings, the required sums are:

$$\sum_i \tilde{x}_{i1}^2 = 9 + 1 + 9 + 1 = 20, \qquad \sum_i \tilde{x}_{i2}^2 = 1 + 9 + 1 + 9 = 20,$$

$$\sum_i \tilde{x}_{i1}\tilde{x}_{i2} = 3 + 3 + 3 + 3 = 12.$$

Divide by $n = 4$:

$$\boxed{\boldsymbol{S} = \frac{1}{4}\begin{bmatrix} 20 & 12 \\ 12 & 20 \end{bmatrix} = \begin{bmatrix} 5 & 3 \\ 3 & 5 \end{bmatrix}}.$$

The positive off-diagonal value 3 shows that the two sensors tend to rise and fall together. This makes a shared PCA direction useful for summarizing their joint variation.

Solution 1(c): Eigenvalues

(c) Find the eigenvalues from the covariance matrix. Solve $\det(\boldsymbol{S} - \lambda \boldsymbol{I}) = 0$:

$$\begin{vmatrix} 5-\lambda & 3 \\ 3 & 5-\lambda \end{vmatrix} = 0 \quad \Longrightarrow \quad (5-\lambda)^2 - 9 = 0.$$

Thus $5 - \lambda = \pm 3$, giving:

$$\boxed{\lambda_1 = 8, \qquad \lambda_2 = 2}.$$

The eigenvalues are variances along their corresponding principal directions. Since $8 > 2$, the first eigenvector will be PC1.

Solution 1(c): Eigenvectors

(c) Find unit eigenvectors and identify PC1. For $\lambda_1 = 8$, the eigenvector equation gives $v_1 = v_2$. For $\lambda_2 = 2$, it gives $v_1 = -v_2$. Normalize both vectors to unit length:

$$\boxed{\boldsymbol{v}_1 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \qquad \boldsymbol{v}_2 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -1 \end{bmatrix}.}$$

Therefore $\boldsymbol{v}_1$ is the first principal direction.
PC1 captures joint changes in both sensors, whereas PC2 captures their differences. The direction signs can be reversed without changing the subspaces or the information retained.

Solution 1(d): Projection and variance

(d) Compute the explained variance and the first score. Total variance is $\lambda_1 + \lambda_2 = 10$, so:

$$\boxed{\mathrm{EVR}(1) = \frac{8}{8+2} = 0.80 = 80\%}.$$

The centered first reading is $\tilde{\boldsymbol{x}}_1 = (3, 1)^{\mathsf{T}}$. Its PC1 coordinate is:

$$z_1 = \boldsymbol{v}_1^{\mathsf{T}} \tilde{\boldsymbol{x}}_1 = \frac{3+1}{\sqrt{2}} = \boxed{2\sqrt{2} \approx 2.828}.$$

The positive score $2\sqrt{2}$ means that the first reading is above the training mean along the shared-sensor direction. One number summarizes their joint change; retaining PC2 would also describe how the two sensors differ.

Task 2: Reconstruction and residuals

A three-channel monitoring system was fitted using training mean:

$$\bar{\boldsymbol{x}} = (10, 20, 30)^{\mathsf{T}}.$$

The covariance eigenvalues and unit directions are:

$$\lambda_1 = 9, \quad \lambda_2 = 2, \quad \lambda_3 = 1,$$

$$\boldsymbol{v}_1 = \frac{1}{\sqrt{2}}(1, 0, 1)^{\mathsf{T}}, \quad \boldsymbol{v}_2 = (0, 1, 0)^{\mathsf{T}}, \quad \boldsymbol{v}_3 = \frac{1}{\sqrt{2}}(1, 0, -1)^{\mathsf{T}}.$$

For a new observation $\boldsymbol{x}_* = (15, 21, 31)^{\mathsf{T}}$:

(a) Choose the smallest r that retains at least 90% variance. Give the training-set average squared reconstruction error for that r.

(b) Compute the retained PCA coordinates of $\boldsymbol{x}_*$.

(c) Reconstruct $\boldsymbol{x}_*$ and compute its squared residual.

(d) Using the threshold $\|\boldsymbol{e}_*\|_2^2 > 4$, determine whether $\boldsymbol{x}_*$ should be flagged.

Solution 2(a): Dimension selection

(a) Select r using cumulative explained variance. The total variance is $9 + 2 + 1 = 12$.

$$\mathrm{EVR}(1) = \frac{9}{12} = 75\%, \qquad \mathrm{EVR}(2) = \frac{9+2}{12} = \boxed{91.7\%}.$$

Therefore the smallest choice meeting 90% is:

$$\boxed{r = 2}.$$

The remaining eigenvalue is $\lambda_3 = 1$. Under the $1/n$ covariance convention, the *average over training observations* of squared reconstruction error is:

$$\boxed{\text{mean training squared error} = \lambda_3 = 1}.$$

This average does not imply that every individual reading has error 1.

Solution 2(b): New-point coordinates

(b) Center and encode the new observation. Use the *training* mean:

$$\tilde{\boldsymbol{x}}_* = \boldsymbol{x}_* - \bar{\boldsymbol{x}} = (15, 21, 31)^\mathsf{T} - (10, 20, 30)^\mathsf{T} = \boxed{(5, 1, 1)^\mathsf{T}}.$$

Project onto the two retained directions:

$$z_1 = \boldsymbol{v}_1^\mathsf{T} \tilde{\boldsymbol{x}}_* = \frac{5+1}{\sqrt{2}} = 3\sqrt{2}, \qquad z_2 = \boldsymbol{v}_2^\mathsf{T} \tilde{\boldsymbol{x}}_* = 1.$$

Therefore the two-dimensional PCA representation is:

$$\boxed{\boldsymbol{z}_* = (3\sqrt{2}, 1)^\mathsf{T}}.$$

Solution 2(c): Reconstruction

(c) Reconstruct from the two retained coordinates. For centered data, $\hat{\tilde{\boldsymbol{x}}}_* = z_1\boldsymbol{v}_1 + z_2\boldsymbol{v}_2$:

$$\hat{\tilde{\boldsymbol{x}}}_* = 3\sqrt{2}\frac{1}{\sqrt{2}}\begin{bmatrix}1\\0\\1\end{bmatrix} + 1\begin{bmatrix}0\\1\\0\end{bmatrix} = \boxed{\begin{bmatrix}3\\1\\3\end{bmatrix}}.$$

Add the training mean back:

$$\boxed{\hat{\boldsymbol{x}}_* = (13, 21, 33)^\mathsf{T}}.$$

The two retained directions cannot reconstruct the component along discarded PC3.

Solution 2(c): Reconstruction error

(c) Compute the squared residual. The residual is the observed reading minus its reconstruction:

$$\boldsymbol{e}_* = \boldsymbol{x}_* - \hat{\boldsymbol{x}}_* = \begin{bmatrix} 15 \\ 21 \\ 31 \end{bmatrix} - \begin{bmatrix} 13 \\ 21 \\ 33 \end{bmatrix} = \boxed{\begin{bmatrix} 2 \\ 0 \\ -2 \end{bmatrix}}.$$

Its squared length is:

$$\boxed{\|\boldsymbol{e}_*\|_2^2 = 2^2 + 0^2 + (-2)^2 = 8}.$$

The error comes from a difference between the first and third channels that the retained subspace cannot represent.

Solution 2(d): Screening decision

(d) Apply the stated residual threshold. The stated flagging rule is:

$$\text{flag if } \|\boldsymbol{e}_*\|_2^2 > 4.$$

Our new observation has:

$$\boxed{8 > 4 \quad \Longrightarrow \quad \text{flag the observation}}.$$

The sample error 8 is also much larger than the *training average* 1 computed in (a).

- The reading departs substantially from the learned two-dimensional pattern.
- This is a screening signal, **not** proof of a sensor failure.
- In practice, set and validate the threshold using representative normal data.

Task 3: SVD and data compression

A centered matrix $\boldsymbol{X}_c \in \mathbb{R}^{8\times 4}$ has a rank-two thin SVD:

$$\boldsymbol{X}_c = \boldsymbol{U}\boldsymbol{D}\boldsymbol{V}^{\mathsf{T}}, \quad \boldsymbol{D} = \begin{bmatrix} 8 & 0 \\ 0 & 4 \end{bmatrix}, \quad \boldsymbol{V} = [\boldsymbol{v}_1, \boldsymbol{v}_2],$$

$$\boldsymbol{v}_1 = \frac{1}{2}(1,1,1,1)^{\mathsf{T}}, \qquad \boldsymbol{v}_2 = \frac{1}{2}(1,1,-1,-1)^{\mathsf{T}}.$$

Here $\boldsymbol{U}^{\mathsf{T}}\boldsymbol{U} = \boldsymbol{I}_2$ and $n = 8$.

(a) Derive the covariance representation $\boldsymbol{S} = \boldsymbol{V}\boldsymbol{D}^2\boldsymbol{V}^{\mathsf{T}}/8$.

(b) Find the nonzero covariance eigenvalues, identify PC1, and compute its explained-variance ratio.

(c) For rank-one reconstruction, compute total squared Frobenius error and average squared error per observation.

(d) Compare storing 8×4 raw values with the rank-one factors and a four-value mean. Compute the storage reduction.

Solution 3(a): SVD and covariance

(a) Derive the covariance identity. For centered observations stored as rows:

$$\boldsymbol{S} = \frac{1}{8}\boldsymbol{X}_c^{\mathsf{T}}\boldsymbol{X}_c.$$

Substitute $\boldsymbol{X}_c = \boldsymbol{U}\boldsymbol{D}\boldsymbol{V}^{\mathsf{T}}$:

$$\begin{aligned}\boldsymbol{S} &= \frac{1}{8}(\boldsymbol{U}\boldsymbol{D}\boldsymbol{V}^{\mathsf{T}})^{\mathsf{T}}(\boldsymbol{U}\boldsymbol{D}\boldsymbol{V}^{\mathsf{T}}) \\ &= \frac{1}{8}\boldsymbol{V}\boldsymbol{D}\boldsymbol{U}^{\mathsf{T}}\boldsymbol{U}\boldsymbol{D}\boldsymbol{V}^{\mathsf{T}}.\end{aligned}$$

Since $\boldsymbol{U}^{\mathsf{T}}\boldsymbol{U} = \boldsymbol{I}_2$:

$$\boxed{\boldsymbol{S} = \boldsymbol{V}\frac{\boldsymbol{D}^2}{8}\boldsymbol{V}^{\mathsf{T}}}.$$

This is why SVD can compute PCA without forming the covariance matrix explicitly.

Solution 3(b): Principal directions

(b) Recover covariance eigenpairs from the SVD. The diagonal entries of $\boldsymbol{D}$ are singular values $s_1 = 8$ and $s_2 = 4$. Therefore:

$$\boxed{\lambda_1 = \frac{8^2}{8} = 8, \qquad \lambda_2 = \frac{4^2}{8} = 2}.$$

There are also two zero eigenvalues because the matrix has rank two. The corresponding nonzero-variance principal directions are the columns of $\boldsymbol{V}$:

$$\boxed{\text{PC1} = \boldsymbol{v}_1 = \frac{1}{2}(1, 1, 1, 1)^{\mathsf{T}}}.$$

The first direction represents coordinated variation across all four features.

Solution 3(b): Explained variance

(b) Compute the retained fraction of variance. The total variance is $8 + 2 + 0 + 0 = 10$. Thus the first principal component explains:

$$\boxed{\mathrm{EVR}(1) = \frac{8}{10} = 0.80 = 80\%}.$$

- Rank one keeps the dominant shared direction.
- The second nonzero component contains the remaining 20% of variance.

In a compression task, choose the retained rank by balancing the variance preserved against the reconstruction error the application can tolerate.

Solution 3(c): Low-rank error

(c) Compute rank-one reconstruction error. A rank-one truncated SVD retains $s_1 = 8$ and discards $s_2 = 4$. By the optimal low-rank approximation result:

$$\|\boldsymbol{X}_c - \boldsymbol{X}_{c,1}\|_F^2 = \sum_{j>1} s_j^2 = s_2^2.$$

Therefore the *total* squared reconstruction error is:

$$\boxed{\|\boldsymbol{X}_c - \boldsymbol{X}_{c,1}\|_F^2 = 16}.$$

For $n = 8$ observations, the average squared error per observation is:

$$\boxed{\frac{16}{8} = 2 = \lambda_2}.$$

This gives the same loss as the discarded covariance variance.

Solution 3(d): Storage cost

(d) Count numbers needed for compression. Storing the original 8×4 data uses:

$$N_{\text{raw}} = 8(4) = \boxed{32 \text{ values}}.$$

For a rank-one approximation, store:

- one left singular vector: 8 values;
- one singular value: 1 value;
- one right singular vector: 4 values;
- the training mean to restore the original scale: 4 values.

Hence:

$$N_{\text{compressed}} = 8 + 1 + 4 + 4 = \boxed{17 \text{ values}}.$$

The reduction is:

$$\boxed{\frac{32 - 17}{32} \times 100\% = 46.875\%}.$$

This saves storage but introduces the reconstruction loss found in (c).

Task 4: PCA before classification

A labeled dataset has two features: x_1 is a variable background reading and x_2 is a small diagnostic signal.

Class y	Observation 1	Observation 2	Observation 3
0	$(-4, -1)$	$(0, -1)$	$(4, -1)$
1	$(-4, +1)$	$(0, +1)$	$(4, +1)$

Use $\boldsymbol{S} = \frac{1}{6} \sum_i \tilde{\boldsymbol{x}}_i \tilde{\boldsymbol{x}}_i^{\mathsf{T}}$.

(a) Compute the covariance matrix, PC1 direction, and PC1 explained-variance ratio.

(b) Project all six observations onto PC1. Can PC1 distinguish the classes?

(c) Compare the best classification accuracy using only PC1 with the rule "predict 1 if PC2 > 0."

(d) A 90% EVR rule keeps one component. Explain which dimensions to validate for classification and how to avoid test leakage.

Solution 4(a): Covariance and PC1

(a) Compute covariance and explained variance. The feature mean is $(0,0)^\mathsf{T}$, so the given observations are already centered.

$$\sum_i x_{i1}^2 = 2(16+0+16) = 64, \qquad \sum_i x_{i2}^2 = 6, \qquad \sum_i x_{i1}x_{i2} = 0.$$

Thus:

$$\boxed{\boldsymbol{S} = \begin{bmatrix} 32/3 & 0 \\ 0 & 1 \end{bmatrix}}.$$

PC1 is $\boldsymbol{v}_1 = (1,0)^\mathsf{T}$ with eigenvalue $\lambda_1 = 32/3$; PC2 is $\boldsymbol{v}_2 = (0,1)^\mathsf{T}$ with $\lambda_2 = 1$.

$$\boxed{\mathrm{EVR}(1) = \frac{32/3}{32/3+1} = \frac{32}{35} \approx 91.4\%}.$$

Solution 4(b): Projected classes

(b) Compute all one-dimensional PC1 coordinates. Because $\boldsymbol{v}_1 = (1, 0)^\mathsf{T}$, the coordinate is $z_1 = \boldsymbol{v}_1^\mathsf{T}\boldsymbol{x} = x_1$.

Class	PC1 score 1	PC1 score 2	PC1 score 3
0	-4	0	4
1	-4	0	4

PC1 alone cannot distinguish the two classes.

Every retained PC1 coordinate occurs once in each class. The feature that carried class information was discarded.

Solution 4(c): Classification accuracy

(c) Compare predictions using PC1 and PC2. PC1 only: for each of $z_1 = -4, 0, 4$, two observations have opposite labels. Any deterministic rule based only on z_1 must get at least one wrong from each pair.

$$\boxed{\text{best accuracy using PC1 only} = \frac{3}{6} = 50\%}.$$

PC2 only: $z_2 = x_2$. Apply the given rule:

$$\hat{y} = \begin{cases} 1, & z_2 > 0, \\ 0, & z_2 \leq 0. \end{cases}$$

All class-0 observations have $z_2 = -1$; all class-1 observations have $z_2 = +1$.

$$\boxed{\text{accuracy using PC2} = \frac{6}{6} = 100\%}.$$

These are accuracies on the six supplied observations, not estimates of future generalization.

Solution 4(d): Model selection

(d) Choose the representation for a real prediction task. A 90% explained-variance rule selects $r = 1$ because PC1 retains 91.4% of total variance. But part (c) shows that discarding PC2 destroys the diagnostic signal in this example. For prediction, compare $r = 1$ and $r = 2$ using validation accuracy rather than EVR alone.

- Split the data before preprocessing. Fit scaling and PCA on training data only.
- Transform validation data with the fitted PCA and use them to choose r.
- Once the pipeline is fixed, evaluate on the held-out test set.

PC1 preserves most input variance here, but PC2 carries the class information. This is why retained variance alone cannot determine the best representation for classification.

Tutorial 8 complete

Questions and discussion